

Yingqi Liu

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EDUCATION

The City College of New York

New York, USA

The Grove School of Engineering, Master of Science

August 2025 – Expected June 2027

- Major: Computer Science
- Relevant Courses: Advanced Algorithms, Computer Vision, Data Privacy
- Cumulative GPA: 3.78

Dalian University of Technology

Dalian, China

(National Demonstration) School of Software Technology, Bachelor of Engineering September 2021 – June 2025

- Major: Software Engineering
- Relevant Courses: Mathematics for Computer Science, Data Structure and Algorithm, Software Engineering
- Cumulative GPA: 3.68

PUBLICATIONS

- W. Lam*, Y. Liu*, C. Di, et al. (*Equal Contribution), “A Single-Pass Approach That Mines Unstructured Robotic Trajectories for Calibrating Surveillance Cameras” *Sensors*, 2026. ([Paper Link](#))

RESEARCH EXPERIENCE

Automated Camera Calibration for Boston Dynamics Spot

New York, USA

Research Assistant, NSF-Funded Project

March 2026 – August 2026

- **Impact:** Supported by NSF Awards [#2131186 \(CISE-MSI\)](#) and [#1827505 \(PFI\)](#), the system significantly enhanced calibration efficiency, laying the groundwork for real-time spatial assistance for blind and low-vision (BLV) individuals. Our results are published on <https://doi.org/10.3390/s26175473>
- **Description:** Engineered an end-to-end automated camera calibration pipeline using a mobile quadruped robot (Boston Dynamics Spot) and LiDAR SLAM. Designed a Virtual Rectangle (ViR) algorithm to dynamically extract geometric structures from unstructured trajectories, achieving focal length errors of approximately 5%, rotational errors of approximately 5° and relative translation errors of approximately 10%.
- **Tools Used:** Python, PyTorch, YOLOv8, Unity, Matplotlib.

Edge-Based Image Recognition System

Dalian, China

Team Leader, Innovation Project

March 2023 – April 2024

- **Impact:** Recognized as a national-level undergraduate innovation project and awarded CNY 5,000 in funding.
- **Description:** Led the team to design and deploy a lightweight image detection algorithm model on the PYNQ-Z2 (FPGA) board. Optimized the deep learning model for edge computing to achieve real-time, hardware-accelerated face detection and mask-wearing determination via a live camera feed.
- **Tools Used:** Python, PyTorch, VHDL, Shell.

HONORS & AWARDS

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| • National Encouragement Scholarship, Ministry of National Education (Top 3%) | December 2024 |
| • Learning Excellence Scholarship, Dalian University of Technology (Top 15%) | December 2024 |
| • National Encouragement Scholarship, Ministry of National Education (Top 3%) | November 2023 |
| • Technology Innovation Award, Dalian University of Technology (Top 4%) | September 2022 |
| • Mathematical Contest In Modeling (MCM) Meritorious Winner, COMAP (Top 7%) | May 2022 |

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, SQL, VHDL
- **Frameworks & Libraries:** PyTorch, YOLOv8, Matplotlib, OpenCV
- **Developer Tools:** Linux/Bash, Git, Unity, FPGA (PYNQ), $\text{\LaTeX}$
- **Languages:** Mandarin Chinese (Native), English (Fluent)